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A unified framework for multi-modal rumor detection via multi-level dynamic interaction with evolving stances

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ABSTRACT

With the escalating dissemination of textual and visual content on the Internet, multi-modal rumor detection has garnered significant scholarly attention in recent research studies. Currently, the prevailing methods in multi-modal rumor detection tend to emphasize the information integration from source posts and images, overlooking the dynamic interaction between multi-modal sources and evolving conversational structures. Furthermore, they fail to recognize the potential advantage that introducing evolving user stances as a form of collective decision-making can improve the model's performance in rumor classification. In this paper, we propose a novel Evolving Stance-aware Dynamic Graph Fusion Network (ESDGFN) to address the above issues. This network aims to integrate the source, the image and the dynamic conversation graph into a unified framework. Specifically, we begin by leveraging a cross-modal transformer for fine-grained feature fusion of the multi-modal sources. Simultaneously, based on the temporal attributes of posts, we construct a set of dynamically changing conversation graphs for each conversation thread, simulating and encoding the evolving stances of users towards the target event within these conversation graphs. Subsequently, we design a multi-level fusion strategy, incorporating both coarse-grained multi-modal feature guidance and fine-grained cross-modal similarity-aware fusion. This strategy aims to generate interactively enhanced multi-modal encoding and dynamic graph representations. The experimental results on both PHEME and Twitter datasets highlight the excellence of our ESDGFN model. It achieves 90.6% accuracy on PHEME, a 3.3% improvement compared to the state-of-the-art method, and 87% accuracy on Twitter, with a 2.4% improvement.

1. Introduction

In today's digital age, the rapid dissemination of information through online platforms such as Weibo and Twitter has fundamentally transformed the way individuals acquire and share information. While this shift has brought many benefits, it has also given rise to an increasingly serious and harmful phenomenon: online rumors. Online rumors refer to false or misleading information spread on the internet, potentially causing negative consequences for the public, including but not limited to decision misguidance, disruption of social stability, and even direct impacts on politics and the economy. For example, false news about Obama's injury triggered drastic fluctuations in the stock market (Kogan, Moskowitz, & Niessner, 2019), and the manipulation of public opinion bots during the U.S. elections (Allcott & Gentzkow, 2017). Given the severity of online rumors, the development of automated rumor detection methods has become crucial.

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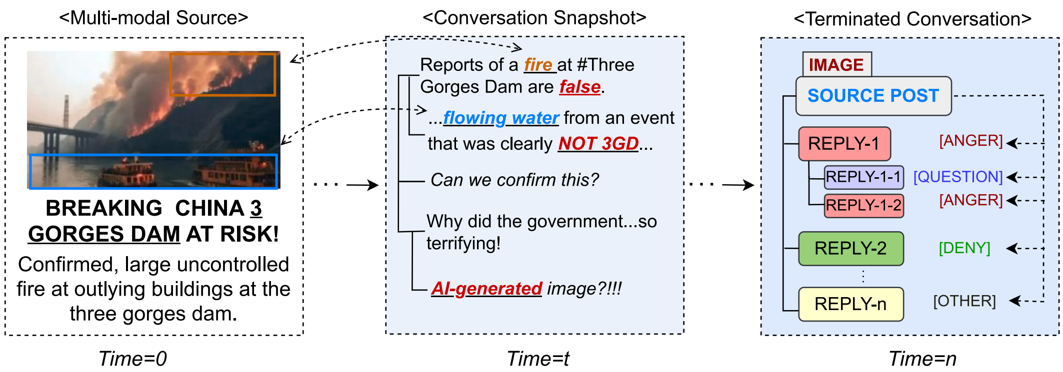

Fig. 1. An evolving rumor case: Three Gorges Dam fire. At time 0, multi-modal information is released. At time t, interactions between comments and multi-modal sources are captured. By time n, the rumor evolution concludes, forming a complete conversation structure with each comment assigned a stance label.

Deep learning has emerged as the dominant paradigm for online rumor detection. Leveraging its outstanding feature learning and pattern recognition capabilities, deep learning models can more accurately capture complex patterns and variations in false information. In the early stages, as rumors primarily existed in textual form, scholars mainly focused on proposing methods centered around text. For instance, methods based on RNNs and CNNs were commonly used to learn the temporal and spatial features of rumor posts (Ma et al., 2016; Yu, Liu, Wu, Wang, Tan, et al., 2017). BERT was employed to uncover deep semantic meanings in rumor texts (Dun, Tu, Chen, Hou, & Yuan, 2021), and so forth.

However, with the continuous advancement of internet information technology, rumor posts have undergone a transformation from being solely textual to adopting a multi-modal format, encompassing text, images, and even videos. This evolution presents a challenge for traditional text-oriented models, as they may not be well-suited for the task of multi-modal rumor detection. Additionally, images can serve as auxiliary cues for identifying rumors, such as: (1) False information often includes sensational or exaggerated images to enhance its spread. (2) Checking if images match the text to spot misinformation. (3) Highlighting false images in conversations as important cues. (4) Providing context that helps understand the text. (5) Acting as the main information source when no text is present. To achieve multi-modal rumor detection, one notable approach involves the fusion of textual and visual features to create a comprehensive representation of rumor data. By integrating visual encoders into detection frameworks, researchers seek to grasp the subtleties conveyed through images (Khattar, Goud, Gupta, & Varma, 2019), enabling a more holistic understanding of the rumor's context and content. Furthermore, techniques such as adversarial learning, as demonstrated in EANN (Wang et al., 2018), have been adopted to enhance the model's ability to discern genuine from fake information. Optical Character Recognition (OCR), as highlighted in Qi et al. (2021), Sun, Qian, Li and Zhu (2023), allows the extraction of textual information from images, further enriching the multi-modal input. To effectively handle the diverse data streams in multi-modal rumor detection, attention mechanisms, as exemplified in MCAN (Wu, Zhan, Zhang, Wang, & Xu, 2021), have been introduced. These mechanisms enable the model to focus on the most salient information within each modality, facilitating improved decision-making and rumor classification.

Although researchers have achieved preliminary success in the field of multi-modal rumor detection, they ignore the following issues. Firstly, existing methods predominantly focus on the feature fusion of multi-modal sources, neglecting the dynamic conversation structure and the comment environment. The dynamic conversation structure provides clues about the spread and evolution of rumors, while the comment environment aids the model in understanding participants' diverse stances and viewpoints, as depicted in Fig. 1. Note that simply concatenating and fusing these features is crude and inefficient. Because not every moment's snapshot in the dynamic conversation graph serves as a key clue, and not every comment provides supporting or refuting opinions. Simple concatenation fusion fails to enable the model to clearly perceive the subtle relationships between different types of features. Therefore, the challenge lies in how to facilitate the model in learning and capturing multi-level, fine-grained interactions between multi-modal sources and dynamic conversation graphs that include follow-up comments.

Furthermore, existing methods do not take into account the evolving group stance as the rumor spreads, or the potential cognitive shifts that may occur, which are crucial factors for the model to analyze rumors. Consider a conversation thread discussing a breaking news event. Initially, users express shock and concern. However, as more information unfolds, some users start questioning the authenticity of the news based on additional facts or context. The conversation evolves with a mix of supportive and skeptical replies. If, at a later stage, the skepticism grows, it might indicate that the source post was misleading, potentially signaling a rumor. Understanding this evolution is crucial for an effective rumor detection model and more detailed discussion is presented in Section 3.6.

To address the aforementioned challenges, we propose a novel Evolving Stance-aware Dynamic Graph Fusion Network (ESDGFN). This network aims to integrate sources, images, and dynamic conversation graphs into a unified framework for multi-modal rumor detection. Specifically, to achieve multi-level interactions between multi-modal sources and dynamic conversation graphs that encompass the comment environment and user stances, we first design a cross-transformer to facilitate fine-grained interaction fusion between source posts and images. Subsequently, leveraging the temporal attributes of posts, we construct a set of dynamically changing conversation graphs for each conversation thread, encoding them with the graph neural network. Regarding the acquisition of user stances, we directly utilize ChatGPT to generate stances for users in the conversation graph at each time step. A gating mechanism is employed to hierarchically fuse user stances and comments, serving as node representations of conversation graphs.

Furthermore, we design a multi-level fusion strategy to produce synergistically enhanced multi-modal sources encoding and dynamic conversation graph representations. On one hand, the learning process of the dynamic conversation graph incorporates coarse-grained guidance from the fusion feature of multi-modal sources. On the other hand, we design a similarity-aware module between the conversation graph and multi-modal sources to diminish attention towards irrelevant comment nodes, resulting in the acquisition of fine-grained source-conversation representations. Finally, in training process, we introduce contrastive learning to align feature distributions between dynamic conversation graphs with the same label.

In summary, the contributions of this paper can be summarized as follows:

- To the best of our knowledge, this is the first study to incorporate source posts, images and dynamic conversation graphs into a unified encoding framework.
- We propose a framework, ESDGFN, for multi-modal rumor detection, which explores fine-grained interactions between source posts and images and comprehensively investigates the multi-level interactions between source features and dynamically evolving conversation graphs.
- We are the first to leverage ChatGPT for generating evolving user stances, thereby aiding the model in understanding variations in user stances, and potential shifts in collective cognition.
- We introduce supervised contrastive learning to mine the potential commonalities and differences between dynamic conversation graphs belonging to different classification labels. Simultaneously, our proposed model outperforms state-of-the-art methods on two public available datasets.

2. Related work

Scholars have adopted diverse methodologies to examine rumors from various angles. In the nascent stages of rumor research, manual techniques such as tracking, screening, and fact-checking held sway. However, with the exponential growth of online information, manual methods proved to be inefficient. Researchers transitioned to automated approaches, embracing techniques such as machine learning (Kwon, Cha, Jung, Chen, & Wang, 2013; Yang, Liu, Yu, & Yang, 2012) and, notably, deep learning, which has emerged as the mainstream method for rumor detection. Based on the type of information utilized, rumor detection can be broadly categorized into single-modal rumor detection and multi-modal rumor detection, which will be discussed in detail below.

2.1. Single-modal rumor detection

In the past, rumor detection research primarily focused on statistical feature analysis, sentiment analysis, and semantic mining of rumor texts. Before the maturity of pre-trained models, researchers commonly utilized traditional neural networks to construct rumor detection frameworks. For instance, Ma et al. (2016) employed the RNN to capture the temporal features among rumor posts. Yu et al. (2017) utilized the CNN to explore potential spatial features. With the popularity of pre-trained models, transformer-based methods, possessing powerful semantic mining capabilities, have been successfully integrated into models. A typical example is the work of Dun et al. (2021), who introduced BERT as a text encoder and leveraged attention mechanisms to learn fine-grained interactions between knowledge entities and post words.

Furthermore, some studies have attempted to integrate text information with other rumor clues (Li, Zhang, & Si, 2019a, 2019b; Ma, Gao, & Wong, 2018a, 2018b, 2019; Shu, Wang, & Liu, 2017; Yang et al., 2020; Yuan, Ma, Zhou, Han, & Hu, 2019). Widely considered is the propagation structure information of rumors. For instance, Bian et al. (2020) used the graph convolutional network to encode the graph structure corresponding to rumor propagation and dispersion. Sun, Zhang, Zheng and Ma (2022) explored dynamic message passing and relevant background knowledge. Data augmentation was introduced by Sun, Qian, Dong, Li, and Zhu (2022) to enhance model robustness. Static stance information is also often integrated into rumor detection frameworks through multi-task learning (Ma, Hu, Ge, & Zhang, 2024; Poddar, Hsu, Lee, & Subramaniyam, 2018; Wei, Xu, & Mao, 2019). Additionally, user characteristics are also frequently incorporated into the model framework from various perspectives (Huang et al., 2022; Lu & Li, 2020; Yuan et al., 2019).

2.2. Multi-modal rumor detection

In the current era of Internet and multimedia technology development, rumors have evolved from being predominantly text-based to incorporating multimedia forms such as images, audio and videos. Traditional text-oriented detection models are evidently inadequate in addressing this situation, giving rise to the emergence of new methods focused on multi-modal features. Currently, multi-modal rumor detection methods can be broadly categorized into two types: optimizing modal fusion and exploring modal inconsistency.

Improving the quality of modal fusion is a focal point for most researchers, rendering it the direction with the richest literature. The main strategy for modal fusion at this stage is to introduce visual information into the model framework and use multi-modal encoders to encode images and text (Jin, Cao, Guo, Zhang, & Luo, 2017). For example, Wang et al. (2018) attempted to extract rumor image features using VGG-19 and employed Text-CNN for text encoding. Khattar et al. (2019) introduced multi-modal reconstruction loss to learn shared features between images and text. Moreover, the success of pre-trained models has injected more possibilities into multi-modal rumor detection. Typically, Yu, Ma, An, and Li (2022) directly utilized BERT and DeiT to encode source posts and images, designing two fusion modules to learn modal interaction information. Jing, Wu, Sun, Fang, and Zhang (2023) extracted spatial and frequency features from images to assist model decision-making. In deeper feature exploration, Qian, Wang, Hu, Fang, and Xu (2021) mined feature differences at different layers of BERT and designed a contextual attention mechanism to investigate modal relationships. Sun, Qian, et al. (2023) delved into dependency relationships between text words and fine-grained correlations with image patches. Zheng et al. (2022) introduced the static multi-modal social graph as supplementary features. In the study of the modal inconsistency, Qi et al. (2021) extracted entities from text and images, assisting the model in judging the relevance between cross-modal entities. Additionally, Sun, Zhang, et al. (2023) proposed a dual-consistency network to explore the consistency on both cross-modal and content knowledge levels.

In comparison to the aforementioned methods, our proposed model not only learns fine-grained interactions among multi-modal information sources, but also integrates the dynamic propagation graph features of rumors and the evolving stances of users into a unified encoding framework. This contributes to the model's understanding of the evolution patterns of rumors, user emotional tendencies, and cognitive transitions.

3. Methodology

3.1. Research objectives

In addressing the challenges discussed in Section 1, our first research objective is to fill the gaps in existing methods concerning the nuanced modeling of interactions between multi-modal sources and dynamic conversation graphs. We achieve this by designing a multi-level fusion strategy that involves coarse-grained multi-modal source feature guidance and fine-grained cross-modal similarity-aware fusion. This strategy aims to obtain interactively enhanced multi-modal features and conversation graph representations.

Furthermore, to enhance the representation of conversation graph nodes and enable the model to grasp the evolving stances of users, our second research objective is to automatically generate user stance labels of comments towards the target event. We employ a gating mechanism to fuse comment nodes and stance features, thereby providing the conversation graph with hierarchical representations.

By combining these two strands of information, our work aims to achieve effective multi-modal rumor detection on social networks.

3.2. Task definition

Multi-modal rumor detection is typically defined as a multi-class classification task, where a classifier is trained through supervised learning by mapping training event instances to their corresponding labels. Specifically, we represent the set of event instances as $E = \{e_1, e_2, \dots, e_k\}$, where e_i is the ith event instance and k is the number of instances. Note that, unlike previous methods that only included source posts and images information, each event instance in this paper contains richer information. Formally, each instance $e = (y, S, I, G_1, G_2, \dots, G_n)$, where $\{G_1, G_2, \dots, G_n\}$ represents a collection of conversation graph snapshots with comments, and S and I are the source post and image, respectively. The ground-truth label y is the target to be mapped, and the number of categories to which y belongs may vary in different types of datasets. For example, in PHEME, y may belong to Rumor (R) or Non-rumor (N). In Twitter datasets, y may belong to Non-rumor (NR), False rumor (FR), True rumor (TR), or Unverified rumor (UR). Ultimately, our goal is to train a model $f(\cdot)$ to successfully classify the event instances in the test set into their corresponding labels.

3.3. Overview

In this section, we will provide a detailed description of our proposed ESDGFN method. As illustrated in Fig. 2, ESDGFN consists of five modules: (1) *Multi-modal sources fusion*. The cross-transformer is designed to facilitate fine-grained interaction and fusion between textual and visual information. (2) *Dynamic conversation graph encoding*. This module focuses on learning the evolution patterns of rumors and the trends of cognitive aspects in comments with reply structures. (3) *Dynamic user stance fusion*. Stance features generated by ChatGPT are integrated into the conversation graph nodes to form the hierarchical representation. (4) *Multi-level feature interaction*. Interactions and adjustments between multi-modal source features and dynamic conversation graph representations are performed to alleviate noise. (5) *Output module*. Contrastive learning is introduced to align feature distributions for effective rumor classification in training process.

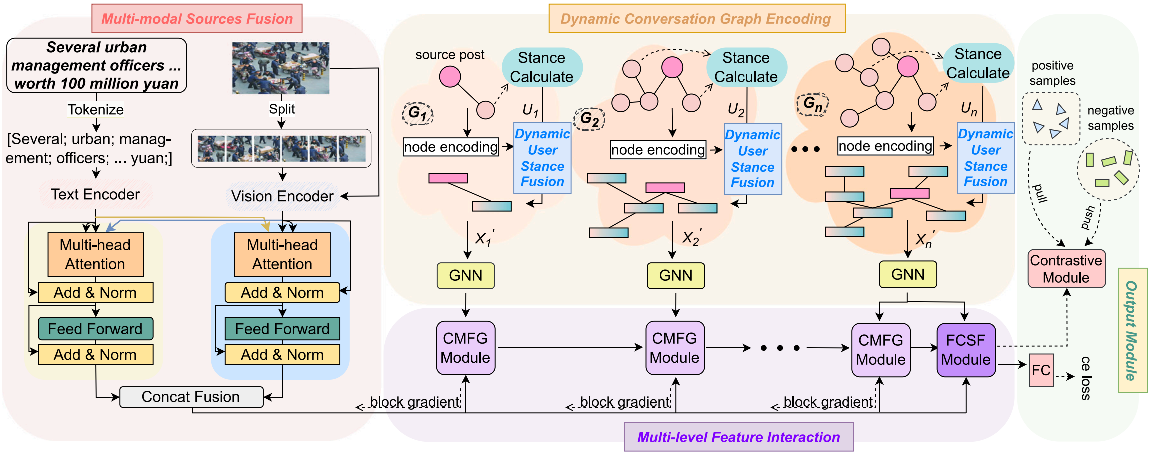

Fig. 2. Overview of our ESDGFN for multi-modal rumor detection.

3.4. Multi-modal sources fusion

Multi-modal sources serve as the primary carriers of rumors (as illustrated at time=0 in Fig. 1) and have been the focal point of most research efforts. There exists fine-grained connectivity between source posts and images. To capture such fine-grained clues, we first attempt to segment text and images into granular tokens and patches, respectively. These are then separately fed into the textual encoder BERT (Devlin, Chang, Lee, & Toutanova, 2019) and visual encoder ViT (Dosovitskiy et al., 2021), to obtain encoded features. Formally, given a source post S with n words and an image I consisting of $m = k \times k$ patches,[1] we represent them as word sequence $s = \{w_i\}_{i=1}^{n}$ and patch sequence $v = \{p_i\}_{i=1}^{m+1}$. Considering the spatial presentation attributes unique to images, the v sequence includes the entire image, ensuring that the model simultaneously takes into account the overall context while learning fine-grained patch details. Subsequently, the s is fed into BERT to obtain the source post feature matrix $X^S = \{x_1^s, x_2^s, x_3^s, \dots, x_n^s\} = BERT(s)$, and similarly, v is input to ViT to obtain the image feature matrix $X^V = \{x_1^v, x_2^v, x_3^v, \dots, x_{m+1}^v\} = ViT(v)$, where the special tokens [CLS], [SEP], and [class] are discarded. To ensure alignment between the feature spaces of the two modalities, we employ projection layers for modality alignment, which can be formalized as $\hat{X}^S = \{s_1, s_2, s_3, \dots, s_n\} = X^S W^S$ and $\hat{X}^V = \{v_1, v_2, v_3, \dots, v_{m+1}\} = X^V W^V$. W is the trainable parameter matrix.

Furthermore, to learn fine-grained feature interactions between the two modalities, we introduce the transformer (Vaswani et al., 2017), a mature framework known for mining long-range dependencies. The transformer block mainly consists of two types of layers: multi-head attention mechanism (MHA) and fully connected feed-forward neural network (FFN). Normalization and residual connection are also used in each layer. Using transformers solely for text features $\hat{X}^S$ or image features $\hat{X}^V$ can only help them mine the correlation of their own modal granularity. Hence, in order to capture the relationship between the two modalities, we improved the attention mechanism, that is, using $\hat{X}^S$ and $\hat{X}^V$ as the query Q of the image-transformer and text-transformer respectively to intervene in fine-grained relationship learning. The formalization process is as follows:

$$Attn(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V, \tag{1}$$

where in text-transformer, $Q = \hat{X}^V$ and $K = V = \hat{X}^S$. On the contrary, $Q = \hat{X}^S$ and $K = V = \hat{X}^V$ in image-transformer. In addition, the calculation process of multi-head attention is as follows:

$$MHA(Q, K, V) = Concat(Attn_1, \dots, Attn_h)W^q, \tag{2}$$

where h is the number of heads and W^q is the parameter matrix. Now, we can obtain the $\hat{X}^{IS} = MHA(\hat{X}^V, \hat{X}^S, \hat{X}^S)$ and $\hat{X}^{IV} = MHA(\hat{X}^S, \hat{X}^V, \hat{X}^V)$. Next, we feed $\hat{X}^{IS}$ and $\hat{X}^{IV}$ into the FFN to obtain $\hat{X}^{IS,FFN} = max(0, \hat{X}^{IS} W_1^{FFN} + b_1)W_2^{FFN}$ and $\hat{X}^{IV,FFN} = max(0, \hat{X}^{IV} W_1^{FFN} + b_1)W_2^{FFN}$ respectively. Finally, we concatenate them to get the feature X^{int} after image–text interaction, which is calculated as:

$$X^{int} = Concat(\hat{X}^{IS,FFN}, \hat{X}^{IV,FFN}), \tag{3}$$

where X^{int} will be used later to perform multi-level interaction with the dynamic conversation graph.

[1] The number of patches into which an image is divided directly influences the amount of information covered. Through testing and evaluation, it has been determined that setting m to 9 or 16 yields optimal results.

3.5. Dynamic conversation graph encoding

Static conversational structure has proven to be a valuable cue feature and has been introduced in many methods (Sun, Qian, et al., 2022). However, most methods only consider the terminated conversation graph representation (as illustrated at time=n in Fig. 1), and they ignore the evolution process of the conversation graph, which can reflect characteristics such as post density (Xia, Xuan, & Yu, 2020) and cognitive shift. In order to learn the dynamics of the rumor conversation graph, we construct a collection of conversation graph snapshots at different times based on the posting time of the posts. Formally, we first assume the dynamic conversation graph $C = \{G_1, G_2, \dots, G_t, \dots, G_n\}$, where G_t represents a snapshot of the conversation structure at time t. Then, the graph neural network is employed to generate the representation of the conversation graph at each moment, which is later used for multi-level interaction with multi-modal sources. Below we will take G_t as an example to explain the graph construction and encoding process in detail.

Construction. G_t is a tree-like structured graph where the root node represents the source post, and the other nodes correspond to follow-up comments. The edges in G_t are formed based on the reply relationships between the posts. As the graph nodes consist of text, we directly utilize BERT to generate corresponding feature for each graph node, thereby obtaining the feature matrix X_t of G_t. Simultaneously, based on the edge relationships in G_t, we can construct the corresponding adjacency matrix A_t. It is important to note that we do not consider the spreading direction of the source post. Although Bian et al. (2020) have explored the impact of different edge directions on the model, through testing, we have not found any significant influence on prediction accuracy. Therefore, this paper only discusses the case where the connections between nodes are undirected. Additionally, each graph node is connected to itself, and the corresponding adjacency matrix A'_t is updated as follows:

$$A'_t = A_t + I, \tag{4}$$

where I is the diagonal matrix. Then, A'_t and X_t will be fed to the graph neural network.

Encoding. Graph Convolutional Network (GCN) (Kipf & Welling, 2017) is one of the most popular graph neural networks. It is characterized by its simplicity and efficiency, modeling relationships between nodes through an adjacency matrix and providing a better receptive field than traditional CNNs. In this study, we consider a two-layer graph convolution, where the adjacency matrix A'_t and the feature matrix X_t of G_t are inputs. The formalized process is as follows:

$$H_t^{(1)} = \sigma(\hat{A}'_t X_t W^{(0)}), \tag{5}$$

where $H_t^{(1)}$ is the first layer output of GCN. The σ is ReLU activation function. $\hat{A}'_t = D^{-\frac{1}{2}} A'_t D^{-\frac{1}{2}}$ is the normalized adjacency matrix, where D is the degree matrix of A'_t. Then, $H_t^{(1)}$ will undergo another graph convolutional layer.

$$H_t^{(2)} = \sigma(\hat{A}_t H_t^{(1)} W^{(1)}), \tag{6}$$

where $H_t^{(2)}$ is the second layer output of GCN.

3.6. Dynamic user stance fusion

The integration and utilization of static stance information have been demonstrated as effective in some existing methods. However, there are still limitations: (a) Existing methods often use stance labels that are relatively simplistic and coarse, failing to capture more nuanced user tendencies and emotional expressions. For example, skepticism carries a stronger sense of doubt or disbelief than a mere question, and denial is more assertive and definitive than skepticism. Additionally, capturing indicators of user emotional states can contribute to classification decisions, as rumors often excel at provoking emotional fluctuations in a crowd. Therefore, enriching the variety of stance labels would aid the model in perceiving more nuanced user expressions. (b) Static stance information fails to reflect the cognitive evolution of user groups or potential shifts. In common scenarios, early participants in rumor discussions are initially drawn in by the content, expressing emotions and support. As the rumor spreads, expressions of skepticism gradually emerge, followed by an increase in users questioning and providing evidence to refute the rumor. Static stances fail to capture these dynamic shifts in user group cognition.

To address issue (a), traditional human-annotated strategies have gradually become inefficient in the era of large models. Hence, we propose a novel approach leveraging the popular ChatGPT to annotate user stances in the conversation graph at each time step. The refined stance labels include support, deny, question, skepticism, neutral, anger, excitement, and irrelevance. More specifically, we treat ChatGPT as a stance annotator, initializing it with prior information on reply relationships in the form of {id: [number], post: [content], in_reply_to_status_id: [number]}. Following this, we simulate the sequential progression of conversation threads by referencing real-world human commenting behavior, sequentially feeding the comments within the discussion thread to ChatGPT based on the chronological order of their posting times. Leveraging its contextual memory capability, ChatGPT generates the stance towards the target event instance for each comment one by one in multi-turn conversations. Regarding the prompt input for ChatGPT, we adopt a straightforward strategy: "What is the stance of the post [content] towards the source post [content]? Please select from {support, …, excitement, irrelevance}".

To address issue (b), we attempt to integrate stance features into the conversation graph at different time steps. Considering that user stances and user comments share the same propagation structure, we directly incorporate user stances into the comment nodes of the conversation graph to form hierarchical node features. Formally, using G_t as an example, a gating mechanism is introduced

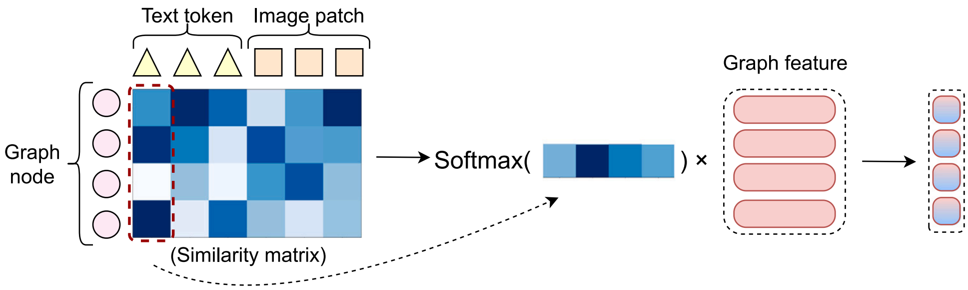

Fig. 3. A visual example of the FCSF module.

to integrate node features X_t and stance features U_t (encoded by BERT). The computation process of the updated hierarchical node features is as follows:

$$X'_t = g_v X_t + (1 - g_v)U_t, \tag{7}$$

where $g_v = sigmoid(W_{gate}[X_t, U_t])$. Eq. (7) can enable the model to weigh the importance of the comment node features and the stance features at each time step, thereby ensuring that the most relevant information is utilized for predictions. If g_v is close to 1, the model relies more on the comment node features X_t. Conversely, if g_v is close to 0, the stance features U_t are given more importance. This adaptability allows the model to incorporate stance information when it is deemed significant and to rely on the inherent comment node features otherwise.

Next, we input the updated feature matrix X'_t and the adjusted adjacency matrix A'_t into Eqs. (5) and (6), respectively, resulting in a new conversation graph representation H'_t that incorporates integrated user stance information. In the following section, we will elaborate on the interaction process between H'_t and the multi-modal source X^{int}.

3.7. Multi-level feature interaction

Existing multi-modal rumor detection methods mostly focus on solving the fusion problem of multi-modal sources, neglecting the interaction with the comment environment structured with replies, as illustrated in Fig. 1. Some comments exhibit close associations with the source post and images, providing crucial evidence. In this section, we propose a multi-level feature interaction and fusion strategy designed to direct the model's attention to more important information and alleviate noise in the contextual environment. It is worth noting that while the introduced standard GCN in this paper has the ability to aggregate text node features, it treats each node as having equal weight. Its aggregation strategy revolves around merging features of surrounding nodes with each node as the center, making it unable to emphasize the importance of the source post. Additionally, it is not suitable for interacting with multi-modal sources and its primary role is to learn structural features.

The multi-level feature interaction and fusion strategy consist of two main modules: Coarse-grained Multi-modal Feature Guidance (CMFG) and Fine-grained Cross-modal Similarity-aware Fusion (FCSF). We will now provide detailed explanations for each of these modules.

Coarse-grained Multi-modal Feature Guidance. The core idea of the CMFG module is to coarsely regulate the learning of the dynamic conversation graph representation at each time step by using the multi-modal source feature X^{int} as guidance. Specifically, we employ the sigmoid function to scale X^{int} and use it as a weighting factor multiplied with H'_t to assist the model in adaptively emphasizing information from different snapshots of the conversation graph. The formalized process can be expressed as follows:

$$d_t = sigmoid(MeanPool(X^{int})W_{id}) \odot MeanPool(H'_t), \tag{8}$$

where d_t represents the conversation graph features under regulation at time t. Considering the sequential nature of the conversation graph, the CMFG module is also responsible for the storage and transmission of conversation graph features at each time step. Referring to the recurrent gating mechanism, the calculation process of the feature storage is as follows:

$$\tilde{h}_t = \tanh(d_t W_{ih} + r_t \odot (h_{t-1})W_{hh}), \tag{9}$$

where $r_t = sigmoid(d_t W_{ir} + h_{t-1} W_{hr})$, and h_{t-1} is the output of the CMFG module at time $t-1$. Then we can obtain the output h_t at the current moment t as follows.

$$h_t = (1 - z_t) \odot \tilde{h}_t + z_t \odot h_{t-1}, \tag{10}$$

where $z_t = sigmoid(d_t W_{iz} + h_{t-1} W_{hz})$. Following the completion of the final time step, we obtain the feature representation denoted as $V_{dynamic}$.

Fine-grained Cross-modal Similarity-aware Fusion. The objective of the FCSF module is to enable the model to capture fine-grained interaction relationships between each comment and the tokens of the source post, and image patches. Specifically, for the z tokens/patches from multi-modal sources, we first compute its similarity matrix of all nodes in the conversation graph. The calculation process is as follows:

$$SM = X^{int}(H_n')^T, \tag{11}$$

where H_n' is the representation output of the conversation graph after undergoing two layers of GCN at the last time step. We selectively compute the similarity matrix for conversation graphs that exhibit a complete reply structure, thereby minimizing unnecessary computational overhead. Subsequently, we apply the softmax function to the similarity matrix of the ith multi-modal token/patch. We then multiply the result by the graph node features, obtaining a similarity-aware representation of the conversation graph tailored to the ith multi-modal token/patch. Fig. 3 gives a visualization case, and the calculation process is as follows:

$$F_i^{sm} = softmax(SM_i)H_n', \tag{12}$$

where $1 \leq i \leq z$. Finally we fuse z similarity-aware conversation graph representations into the multi-modal feature X^{int} and obtain V_{modal}, as follows:

$$V_{modal} = MeanPool(X^{int}W^{int} + [F_1^{sm}; \ldots; F_z^{sm}]W^{sm}). \tag{13}$$

where W is the trainable weight matrix. V_{modal} and $V_{dynamic}$ will be merged for final rumor classification.

3.8. Output module

After the enhanced interaction, we directly concatenate $V_{dynamic}$ and V_{modal}, expressed as follows:

$$V_{output} = concat(V_{dynamic}, V_{modal}). \tag{14}$$

Then, we feed V_{output} into a fully connected layer with softmax function to obtain the predicted probability distribution $\hat{y}$, which is calculated as:

$$\hat{y} = softmax(W_o V_{output} + b_o), \tag{15}$$

where W and b are the parameter matrix and bias respectively.

In the model training stage, we introduce contrastive learning (Khosla et al., 2020) to align the feature distribution of dynamic conversation graphs in the same category, and also help to alleviate the problem of weak generalization of cross-entropy loss (Liu, Wen, Yu, & Yang, 2016). Contrastive learning aims to maximize the consistency among positive samples while enlarging the feature distance between negative samples.

In this study, we consider dynamic conversation graph representations with the same label as positive sample pairs, and those with different labels as negative sample pairs. Subsequently, we increase the cosine similarity between positive sample pairs in high-dimensional space while decreasing the vector similarity between negative sample pairs to aid the model in learning invariant representations. Finally, we combine the contrastive loss with the cross-entropy loss to form the ultimate loss function. The formalized process is as follows:

$$\mathcal{L} = \mathcal{L}_{ce} + \alpha \mathcal{L}_{sup}, \tag{16}$$

where

$$\mathcal{L}_{ce} = -\frac{1}{K}\sum_{k=1}^{K}\sum_{c=1}^{M} y_{k,c} log(\hat{y}_{k,c}), \tag{17}$$

$$\mathcal{L}_{sup} = -\sum_{k \in K} log\left\{\frac{1}{|P(k)|}\sum_{p \in P(k)} \frac{exp(sim(V_{dynamic,k}, V_{dynamic,p})\tau)}{\sum_{a \in A(k)} exp(sim(V_{dynamic,k}, V_{dynamic,a})\tau)}\right\}, \tag{18}$$

and α is the adjustable hyperparameter. In $\mathcal{L}_{ce}$, K is the number of samples and M is the number of classes. $y_{k,c}$ is the ground-truth label. In $\mathcal{L}_{sup}$, $P(k)$ is the set of positive samples related to sample k. $V_{dynamic,k}$ and $V_{dynamic,p}$ are the representations of samples k and p respectively. $A(k)$ is the set of negative samples related to sample k. $sim()$ is the cosine similarity function. $\tau \in \mathcal{R}^+$ is the adjustable temperature parameter.

4. Experiments

4.1. Datasets

We evaluate the effectiveness of our ESDGFN method on two publicly available datasets, namely PHEME[2] (Kochkina, Liakata, & Zubiaga, 2018) and Twitter[3] (Ma, Gao, & Wong, 2017), both collected from the popular *twitter* platform. The event instances in the

[2] https://figshare.com/search?q=pheme.

[3] https://www.dropbox.com/s/7ewzdrbelpmrnxu/.

Table 1
Statistics of the datasets.

Dataset	# Posts	# Users	# Sources	# Images	# NR	# FR	# UR	# TR
Twitter	26,912	24,619	1001	1001	331	250	159	261
PHEME	20,934	12,438	1198	1198	599	599	N/A	N/A

PHEME dataset consist only of textual source posts and comments. To meet the requirements of our model for multi-modal input, we crawl corresponding images based on the provided links and retain only those event instances in the dataset that contain both textual and image information. Additionally, PHEME is commonly used for binary classification tasks, with labels indicating whether the instance is a rumor (R) or non-rumor (N). The Twitter is an integration of Twitter15 and Twitter16 datasets (Ma et al., 2017), with the addition of comment information, visual information, and conversation structure information. Twitter is often employed for four-class classification tasks, with labels denoting non-rumor (NR), false rumor (FR), true rumor (TR), and unverified rumor (UR). Similarly, to align with the model's input requirements, we keep only those event instances in the dataset that include multi-modal information. It is important to note that, to ensure fairness, all baselines, including single-modal methods, are trained and tested using datasets of the same size in subsequent comparative experiments. The detailed statistical results are presented in Table 1. Additionally, we utilize ChatGPT to extract stance information for each comment. The overall distribution of stance labels in the PHEME and Twitter datasets is illustrated in Fig. 4.

4.2. Baselines

We compare our method with the following state-of-the-art baselines.
Single-modal based methods:

- BERT (Devlin et al., 2019) is a robust text-based model built upon the Transformer architecture.
- ViT (Dosovitskiy et al., 2021) is a popular and powerful vision-based model, which has a similar structure to BERT and is often used for image semantic mining.
- BiGCN[4] (Bian et al., 2020) designs a root enhancement strategy to increase the weight of source posts in the conversation graph, and also explores the impact of the rumor propagation direction on the model.
- DDGCN[5] (Sun, Zhang, et al., 2022) designs a dual dynamic encoder to integrate dynamically delivered messages and background knowledge into a unified detection framework.

Multi-modal based methods:

- EANN[6] (Wang et al., 2018) introduces VGG-19 to encode image features and employs Text-CNN to mine text information.
- MVAE (Khattar et al., 2019) utilizes the idea of multi-modal variational autoencoders. It reconstructs image and text modalities and then discovers the correlation between modalities.
- HMCAN[7] (Qian et al., 2021) proposes a contextual attention network that not only utilizes multi-modal information but also learns a complete hierarchical representation of text.
- CLIP (Radford et al., 2021) is a classic pre-trained multi-modal model based on contrastive learning technology.
- BLIP-2 (Li, Li, Savarese, & Hoi, 2023) is a novel pre-trained multi-modal model known for its lightweight characteristics.
- MFAN[8] (Zheng et al., 2022) designs the cross-modal attention mechanisms to mine the correlation between images and texts, and additionally introduces static social graphs as auxiliary features.
- MGIN (Sun, Qian, et al., 2023) utilizes the dependencies between text words, but also constructs a cross-modal graph to mine fine-grained relationships between modalities.

[4] https://github.com/TianBian95/BiGCN
[5] https://github.com/MengzSun/DDGCN
[6] https://github.com/yaqingwang/EANN-KDD18
[7] https://github.com/wangjinguang502/HMCAN
[8] https://github.com/drivsaf/MFAN

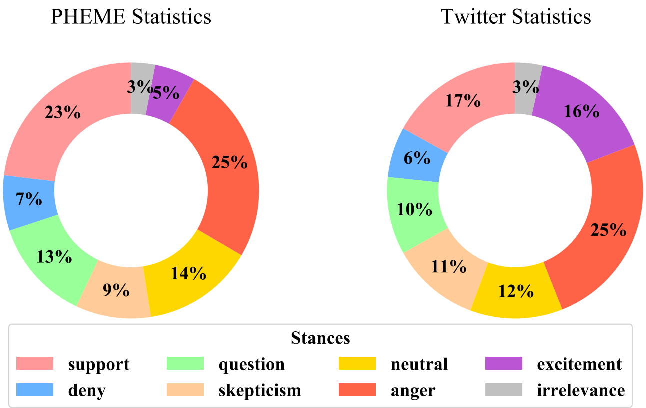

Fig. 4. Statistics of stance labels.

Table 2
Training time overhead of various models (minutes).

Method	PHEME	Twitter
ViT	9.5	7.5
BERT	5.6	5.7
BiGCN	17.8	28.7
DDGCN	25.5	26.0
EANN	12.3	8.8
MVAE	13.2	8.6
HMCAN	10.7	14.4
CLIP	8.2	10.0
BLIP-2	7.5	11.6
MFAN	24.8	30.0
MGIN	48.5	49.8
ESDGFN	31.5	36.9

4.3. Implementation details

The proposed ESDGFN framework is implemented based on PyTorch (Ketkar, 2017). We employ Accuracy, Precision, Recall and F_1-measure (F_1) as evaluation metrics and adopt five-fold cross-validation. In this study, we set the batch size to 32, learning rate to 0.0001. The parameter h in Eq. (2) is configured as 8. The α in Eq. (16) is set to 0.01, and τ in Eq. (18) is set to 1.5. In order to prevent overfitting, we set early stopping with 10 patience and use the Adam algorithm to optimize the objective. To ensure fairness, a uniform parameter configuration is employed across all baselines. At the same time, facing the new multi-modal version of PHEME and Twitter datasets, we use the PyTorch to reproduce all baselines and perform the same five-fold cross-validation as ESDGFN. Furthermore, certain models have unique hyperparameters, which are set according to their original papers. In BiGCN, the $TDdroprate$ and $BUdroprate$ parameters are set to 0.2. In MFAN, λ_c and λ_a in the loss function are set to 2.15 and 1.55 respectively. In MGIN, the number of image patches is set to 16.

The computations of ESDGFN and the baselines are conducted on an NVIDIA GeForce RTX 2080 Ti, using driver version 535.113.01 and CUDA version 12.2. The training time overhead of various models is presented in Table 2. It can be observed that methods involving graph structures and comment information, i.e., BiGCN, DDGCN, MFAN, MGIN, and ESDGFN, often require greater computational overhead. However, it is not necessarily the case that introducing more features or consuming greater computational resources will lead to better prediction results. This may also depend on the encoder, feature fusion strategy, training method, etc., adopted by different methods, as well as the different patterns of data. More specific discussion and analysis can be found in Sections 4.5 and 4.6.

4.4. Time complexity analysis

The time complexity analysis of ESDGFN revolves around its core components, each contributing distinctively to the overall computational overhead. Firstly, the source text encoding primarily relies on attention mechanisms, whose time complexity increases with the length of the text. Therefore, it can be estimated as $O(n^2)$, where n represents the number of words in the text. Similarly, the time complexity for image encoding can be approximated as $O(P^2)$, where P denotes the number of patches in the image. Secondly, during the dynamic interaction phase, each node in the conversation graph at each time step needs to be processed. Therefore, the time complexity for a single time step can be expressed as $O(V)$, where V is the total number of nodes in the conversation graph. Considering the entire dynamic evolution of the conversation thread across multiple time steps (T), the cumulative time

Table 3
Results of various models on PHEME.

Type	Method	Accuracy	Rumor			Non-rumor		
			Precision	Recall	F_1	Precision	Recall	F_1
Single-modal	ViT	0.771	0.760	0.791	0.769	0.783	0.757	0.761
	BERT	0.863	0.885	0.857	0.867	0.849	0.862	0.851
	BiGCN	0.830	0.812	0.855	0.827	0.850	0.807	0.822
	DDGCN	0.867	0.862	0.877	0.862	0.885	0.869	0.867
Multi-modal	EANN	0.776	0.757	0.810	0.776	0.796	0.758	0.768
	MVAE	0.833	0.802	0.888	0.839	0.888	0.775	0.820
	HMCAN	0.866	0.880	0.849	0.862	0.850	0.870	0.858
	CLIP	0.857	0.845	0.857	0.849	0.865	0.852	0.857
	BLIP-2	0.871	0.850	0.895	0.869	0.895	0.845	0.865
	MFAN	0.852	0.834	0.887	0.853	0.879	0.819	0.840
	MGIN	0.876	0.868	0.889	0.874	0.892	0.858	0.870
	ESDGFN	**0.906**	**0.912**	**0.916**	**0.910**	**0.902**	**0.903**	**0.898**

complexity for this phase is $O(T \cdot V)$. Additionally, in other parts of the framework, such as user stance feature fusion, although these operations also contribute to the overall computational load, their time complexity is relatively lower compared to the core components mentioned above. Therefore, they are typically considered negligible in the overall complexity analysis. Finally, during the training phase, the time complexity is closely related to the batch size due to the use of contrastive learning techniques. It can be approximately expressed as $O(B^2)$, where B is the batch size. This complexity primarily arises from the pairwise comparisons or similarity calculations involved in contrastive learning. In summary, the overall time complexity of the ESDGFN framework can be approximated as the sum of the complexities of the aforementioned components, which is $O(n^2 + P^2 + T \cdot V + B^2)$.

4.5. Performance comparison

We conduct a comparative analysis of ESDGFN against other benchmarks on the PHEME and Twitter datasets, presented in Tables 3 and 4. Compared to single-modal methods or multi-modal methods, our ESDGFN achieves state-of-the-art results on most metrics. In particular, we achieve an average improvement of 3.7% on the both datasets compared to the best single-modality baseline DDGCN, and an improvement of 2.9% compared to the strongest multi-modal baseline MGIN. These results confirm the superior performance of our model. Furthermore, we make the following noteworthy observations.

Among single-modal methods, compared to BERT using source post information, ViT, which only considers visual features, lags behind significantly in performance. This is mainly because text is the main information carrier of rumors, while images more often play a supplementary and auxiliary role. BiGCN and DDGCN introduce propagation structure features while considering rumor text. BiGCN places particular emphasis on the directional aspects of propagation, while DDGCN focuses more on the dynamic nature of information spread. However, the notable performance gap between these methods and ESDGFN stems from their insufficient exploration of multi-modal information and their lack of consideration for interactions between the propagation graph and various modalities.

Among multi-modal methods, EANN and MVAE have poor performance due to the use of outdated multi-modal encoders. HMCAN and MGIN, both relying on attention mechanisms, demonstrate commendable performance across both datasets. But their problem lies in the neglect of structural information. Similarly, CLIP and BLIP-2, which adopt multi-modal pretraining strategies, focus on mining the interaction between image and text modalities. Although they rely on powerful coding capabilities to perform well on PHEME, where textual and visual features are prominent. However, because they are not models specifically designed for rumor detection tasks, they lack the ability to comprehensively utilize other key clues, which leads to their weak performance on Twitter. Notably, MFAN introduces a static social graph as auxiliary features, yet its neglect of dynamic modeling for conversation interactions and tracking user stances represents a notable limitation. By contrast, our ESDGFN integrates fine-grained interactions among multi-modal sources, multi-level interactions with dynamic conversation graphs, and the evolution of collective stances into a unified framework. This not only aids the model in adapting to varying feature distributions in the data, as elaborated in Section 4.6, but also enables the model to learn more comprehensive and hierarchical clues to assist in classification decisions. We believe that this integration is a key factor contributing to the optimal performance achieved by our method.

However, it is important to note that during testing on the Twitter dataset, ESDGFN adopts a mixed training approach combining cross-entropy and contrastive loss, which makes the training process sensitive to the ratio of positive and negative samples within a batch. Due to the limited number of samples in the UR class, this directly impacts the performance of ESDGFN in terms of F1 score for the UR class. In the future, we may consider designing some data augmentation strategies to address such data imbalance issues. In addition, although the F1 score of TR is impressive, it lags behind MGIN, possibly due to MGIN's unique MOE training techniques and OCR assistance. However, our ESDGFN significantly outperforms MGIN in all other metrics.

Table 4
Results of various models on Twitter.

Type	Method	Accuracy	NR F_1	FR F_1	TR F_1	UR F_1
Single-modal	ViT	0.674	0.723	0.623	0.697	0.507
	BERT	0.801	0.823	0.802	0.803	0.654
	BiGCN	0.841	0.927	0.741	0.849	0.723
	DDGCN	0.843	0.911	0.751	0.857	0.789
Multi-modal	EANN	0.793	0.806	0.800	0.802	0.648
	MVAE	0.828	0.825	0.827	0.832	0.788
	HMCAN	0.839	0.840	0.819	0.835	0.780
	CLIP	0.767	0.744	0.798	0.775	0.651
	BLIP-2	0.800	0.767	0.820	0.845	0.718
	MFAN	0.833	0.916	0.821	0.826	**0.803**
	MGIN	0.849	0.883	0.793	**0.893**	0.737
	ESDGFN	**0.870**	**0.945**	**0.834**	0.867	0.742

Table 5
Ablation study on PHEME and Twitter.

Model	PHEME		Twitter	
	Accuracy	$avg F_1$	Accuracy	$avg F_1$
ESDGFN	**0.906**	**0.904**	**0.870**	**0.847**
w/o dynamic	0.883	0.880	0.827	0.789
w/o modality	0.848	0.843	0.854	0.809
w/o interaction	0.888	0.887	0.857	0.829
w/o stance	0.893	0.888	0.859	0.833
w/o vision	0.897	0.896	0.860	0.833
w/o cl	0.903	0.900	0.865	0.834
Vision-only	0.799	0.795	0.686	0.653
Text-only	0.871	0.868	0.808	0.779
Graph-only	0.834	0.828	0.847	0.812
Stance-only	0.600	0.580	0.495	0.439
Text+Graph	0.894	0.892	0.854	0.837
Text+Stance	0.889	0.885	0.814	0.803
Vision+Graph	0.832	0.826	0.797	0.758
Vision+Stance	0.795	0.789	0.678	0.641
Vision+Graph+Stance	0.834	0.830	0.807	0.784

4.6. Ablation study

To assess the contributions of different modules in ESDGFN, we compare it with six distinct variants: **w/o dynamic**: We remove the dynamic conversation thread structure. **w/o modality**: We exclude the multi-modal sources fusion module. **w/o interaction**: We eliminate the multi-level feature interaction module. **w/o stance**: User stance information is not utilized. **w/o vision**: Visual information is disregarded. **w/o cl**: Contrastive loss is omitted. The results of the ablation experiments are presented in Table 5, and we have the following observations:

- The PHEME and Twitter datasets manifest distinct data styles. In the PHEME dataset, multi-modal sources are pivotal in uncovering rumors, as evidenced by the noteworthy accuracy decline (∼5.8%) observed upon the exclusion of the multi-modal source fusion module (w/o modality). The difference is that in the Twitter dataset, the dynamic conversation threads furnish the most valuable classification clues. This explains the significant accuracy decline (∼4.3%) observed when omitting the dynamic conversation graph (w/o dynamic).

- The exclusion of the multi-level feature interaction module (w/o interaction) prevents the model from discerning the nuanced interactions between multi-modal sources and dynamic conversation graphs. Consequently, there is a reduction in accuracy by 1.8% and 1.3% on the PHEME and Twitter datasets. Additionally, this module serves the purpose of noise mitigation, facilitating the model in extracting more pertinent features from a fine-grained perspective.

- Omitting user stance information results in a decline in accuracy by 1.3% and 1.1% on the PHEME and Twitter datasets, respectively. As discussed in Sections 1 and 3.6, the spread of rumors involves potential shifts in cognition or stances within user communities. Therefore, learning stance evolution is beneficial for the model.

- When the model does not consider visual features, the accuracy drops by 0.9% and 1% on the PHEME and Twitter datasets respectively. Because images not only carry partial rumor-related information but are also often utilized in spreading misinformation through the deployment of exaggerated visuals. In addition, Table 5 also demonstrates that the combined use of contrast loss and cross-entropy loss significantly enhances the classification ability of the model.

Table 6
Statistics of the Weibo dataset.

Dataset	# Posts	# Users	# Sources	# Images	# NR	# FR	# UR	# TR
Weibo	3,805,656	2,746,818	4,664	N/A	2,351	2,313	N/A	N/A

Table 7
Results of various models on Weibo.

Type	Method	Accuracy	Rumor			Non-rumor		
			Precision	Recall	F_1	Precision	Recall	F_1
Comparison	BERT	0.917	0.917	0.911	0.912	0.915	0.925	0.918
	BiGCN	0.923	0.903	0.952	0.924	0.948	0.899	0.919
	DDGCN	0.932	0.922	0.945	0.932	0.944	0.916	0.928
	ESDGFN	**0.947**	**0.936**	**0.962**	**0.947**	**0.960**	**0.934**	**0.945**
Ablation	w/o dynamic	0.922	0.923	0.918	0.919	0.920	0.928	0.922
	w/o modality	0.930	0.925	0.927	0.923	0.933	0.927	0.928
	w/o interaction	0.934	0.926	0.943	0.932	0.945	0.925	0.933
	w/o stance	0.938	0.926	0.953	0.937	0.954	0.923	0.936
	w/o cl	0.935	0.933	0.940	0.934	0.938	0.931	0.932

To further analyze and validate the roles and interactions of different modalities in the ESDGFN framework, we construct the following variants. Specifically, **Vision-only** uses only image information for classification. **Text-only** uses only the source text information. **Graph-only** considers only the features of the dynamic conversation graph, with node representations fixed as comment features. **Stance-only** uses only stance information for prediction. **Text+Graph** combines source text and conversation graph features for classification. **Text+Stance** combines source text and stance features. **Vision+Graph** combines image and conversation graph features. **Vision+Stance** combines image and stance features. **Vision+Graph+Stance** uses image, conversation graph, and stance information simultaneously. All these experiments use a complementary training strategy of cross-entropy and contrastive learning. The comparative experiment results are shown in Table 5.

From the results of the Vision-only, Text-only, Graph-only, and Stance-only experiments, it is evident that source text and conversation graph features are the most important classification cues, followed by image and stance features. The performance difference between Text-only and Graph-only shows that the PHEME dataset emphasizes text features, while the Twitter dataset, with its deeper conversation structure, emphasizes structural features. This aligns with our previous ablation analysis. Furthermore, in the series of experiments combining features, it can be observed that combining source text with either conversation graph features or stance features improves classification performance. However, combining image features with conversation graph features or stance features has a negative effect. This is primarily because comments and their corresponding stances are generated around the source text, while images typically serve as supplementary information. Forcing images to interact with every comment and stance can create confusion during learning or training, negatively impacting prediction accuracy. Therefore, when interacting with comments and stances, the involvement of source text is necessary. Nevertheless, integrating all these features together can still be beneficial, as each feature can play its role in different scenarios.

4.7. Chinese text-only analysis

The ESDGFN framework is not only applicable to the multi-modal rumor detection task, but it can also be applied to pure text-based rumor detection. To validate the model's transferability in text-based rumor tasks and its generalization on Chinese corpora, we choose the Weibo dataset (Ma et al., 2016) for training and testing. In terms of experimental settings, we set the learning rate to 0.0001, batch size to 32, parameter h in Eq. (2) to 8, parameter α in Eq. (16) to 0.01, and parameter τ in Eq. (18) to 1. Additionally, we select only single-modal methods for comparison. The dataset statistics are presented in Table 6, and the experimental results are shown in Table 7. It can be observed that ESDGFN still achieves excellent prediction results in Chinese dataset, surpassing other baselines. At the same time, from the comparison results of removing dynamic structure (w/o dynamic) and source text (w/o modality), it can be observed that the Weibo dataset does not exhibit prominent classification features like PHEME or Twitter. Instead, it is relatively more balanced. Additionally, the removal of the image module is straightforward, as the text feature and image feature are directly concatenated in ESDGFN. The image–text feature is passed to a complex multi-level feature interaction module subsequently. However, as depicted in Fig. 3, the image–text feature interact in parallel with the graph node feature. Therefore, even if the image patch features are removed, it does not affect the entire encoding framework.

4.8. Stance statistical analysis

Investigating the proportional distribution of various stance labels among users for rumors and non-rumors is crucial for effective rumor identification. Our observations on the PHEME dataset reveal that, for rumor samples, users tend to exhibit a higher proportion of denial stances compared to non-rumors, with an increase of 20%. Similarly, in the Twitter dataset, the proportion of denial stances towards rumors surpasses that towards non-rumors by 45%. This indicates that rumors prompt a higher incidence of users expressing question or denial-type comments. Simultaneously, such expressions become crucial classification cues for the model.

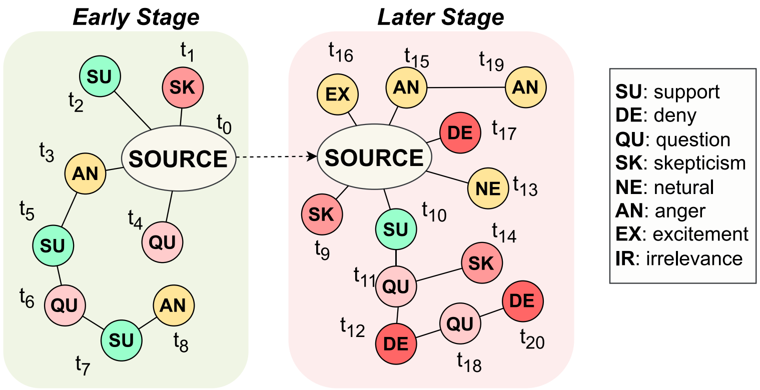

Fig. 5. A rumor case, where the corresponding *SOURCE* is "BREAKING Soldier shot near Canada's parliament" and t represents the posting time.

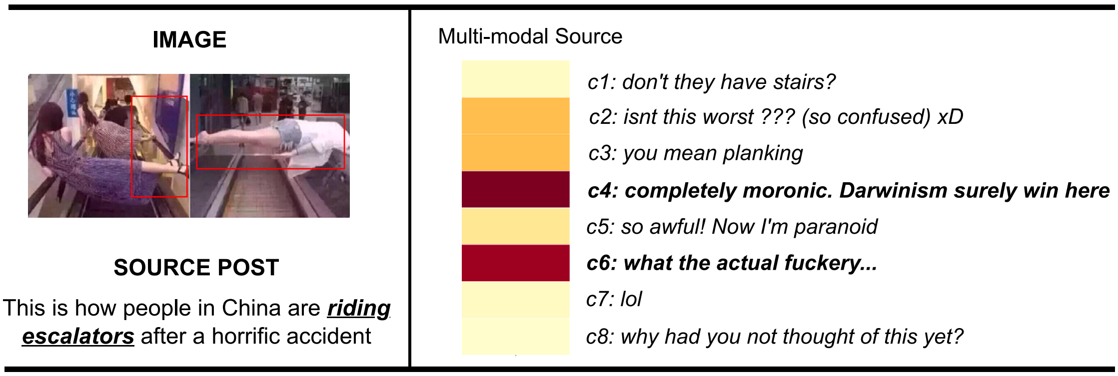

Fig. 6. A multi-modal rumor case, where the attention weight between multi-modal sources (post+image) and comments is visualized, with darker colors representing stronger relevance.

Furthermore, our analysis indicates a notable difference in the expression of anger among users in rumor and non-rumor cases. Specifically, in the PHEME dataset, instances of users displaying angry stances in rumor-related cases are 8% higher compared to non-rumors. In the Twitter dataset, the proportion of anger expressions in rumors exceeds that in non-rumors by 29%. This observation strongly supports the idea that rumors tend to leverage emotional triggers, especially anger, to drive heightened discussion and dissemination. Hence, refining and enriching user stance labels is beneficial for the model to better perceive the difference between rumors and non-rumors.

4.9. Case study

In this section, we will further illustrate the effectiveness of our proposed ESDGFN model through a rumor case that was successfully detected by the model. Fig. 5 presents snapshots of the target event instance during its early and later stages of propagation. In the early stages, user stances towards the event predominantly revolved around support (SU) and question (QU). This indicates that, before the event gained significant traction, the majority of users exhibited curiosity, raised inquiries, or simply engaged in swift browsing and forwarding of messages. Of course, there could also be instances of orchestrated efforts, such as paid commentators artificially boosting the post's popularity.

However, in the later stages of propagation, an increasing number of skeptical and denial expressions emerge, often accompanied by supporting evidence. Additionally, we observed that the reply structure can indicate more nuanced stance relationships among users. For instance, the comment at t_{10} express support (SU) towards the source post, but subsequent replies exhibit predominantly opposing stances (DE, SK and QU). Our ESDGFN model successfully identifies such rumor patterns by grasping such changes in user stances and attitudes as well as the reply relationships between them.

In addition, we also show a typical multi-modal rumor case. From Fig. 6, we can observe that rumors often make full use of exaggerated visuals to trigger discussions, but this also leaves us with classification clues, namely exaggerated visual effects and replies expressing surprise. These clues tend to increase the probability that the source post is a rumor. Furthermore, our model can also perceive the importance of different reviews. For example, in Fig. 6 we can see that there is a stronger correlation between the $c4$ and the multi-modal sources. This ability can help the model focus on more important information and alleviate noise.

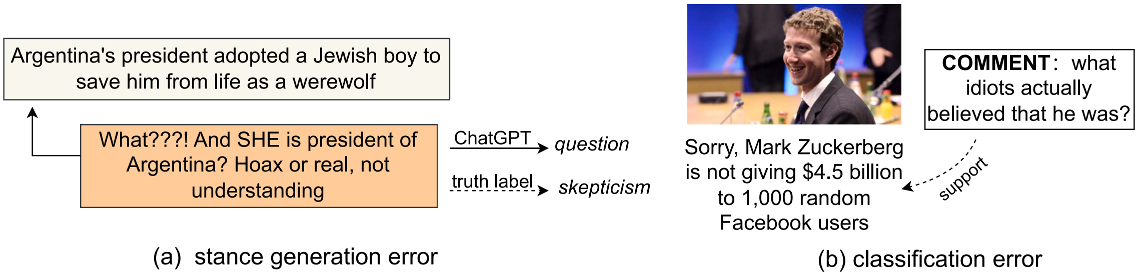

Fig. 7. (a) A case where the incorrect stance is generated. (b) A rumor case misclassified by the ESDGFN model. Note that this conversation thread only contains three replies, and the most informative comment inadvertently supports the rumor conveyed in the source post.

4.10. Error analysis

The ESDGFN is not a perfect multi-modal rumor detection model, and it will also make some mistakes, as illustrated in Fig. 7. When generating user stances based on ChatGPT, it may incur annotation errors due to the instability of ChatGPT. For example, as depicted in Fig. 7(a), it occasionally categorizes the post with question marks blindly as *question*, but upon deeper semantic exploration, it becomes evident that the post should be classified as *skepticism*. Faced with this limitation, we plan to consider incorporating more advanced models such as GPT-4 in the future to enhance annotation accuracy.

In addition, as depicted in Fig. 7(b), ESDGFN is prone to misclassifying rumor events as non-rumors when faced with a lack of propagation structure information and misleading indicative comments. To address such challenges, we are considering the introduction of a news environment. This involves clustering other conversation threads associated with the same event. By doing so, we aim to provide the model with a more comprehensive perspective and contextual knowledge, mitigating the risk of misjudgments caused by local disturbances.

5. Conclusion

5.1. Theoretical implications

In this paper, we integrate source posts, images and dynamic conversation graphs into a unified framework and propose a multi-modal rumor detection model named ESDGFN. Specifically, we first design a cross-transformer to mine fine-grained interactions between source posts and images, and then construct and encode dynamic conversation thread graphs. At the same time, we propose a multi-level interaction strategy to explore the complex interactions between multi-modal sources and dynamic conversation graphs. In addition, we also use ChatGPT to generate the stance information of the topic participants and integrated it into the characteristics of each node in the conversation graph to help the model grasp the changing rules of the user's stances. Results on two publicly available datasets show that our method is competitive and outperforms SOTAs. In the future, we will pay attention to the introduction of external knowledge and the development of early rumor detection models.

5.2. Practical implications

In the era of advanced multimedia and the Internet, it is almost impossible to manually complete fact-checking one by one. Therefore, it is necessary to design an automated multi-modal rumor detection framework to complete the review and screening of messages. Especially for review agencies, multi-modal rumor detection models with high recall rates can first screen out low-authenticity news as much as possible, and then conduct a second round of review by staff, which greatly improves the efficiency of rumor classification. In addition, this study also demonstrates the importance of unifying multi-modal source features and propagation features, especially in the face of different styles of false news (discussed in Section 4.6). In future work, we will further optimize the rumor detection system and provide mature solutions for model deployment.

CRediT authorship contribution statement

Tiening Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Chengwei Liu:** Supervision, Data curation. **Lizhi Chen:** Supervision, Project administration, Data curation. **Zhong Qian:** Supervision, Project administration, Data curation. **Peifeng Li:** Writing – review & editing, Supervision, Project administration. **Qiaoming Zhu:** Writing – review & editing, Supervision.

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Data availability

Data will be made available on request.

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